

VIT®

Vellore Institute of Technology

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School of Computer Science and Engineering

Fall Semester 2025-26

BCSE331P – Exploratory Data Analysis

Lab Assessment 4

Bivariate Analysis Using R Programming

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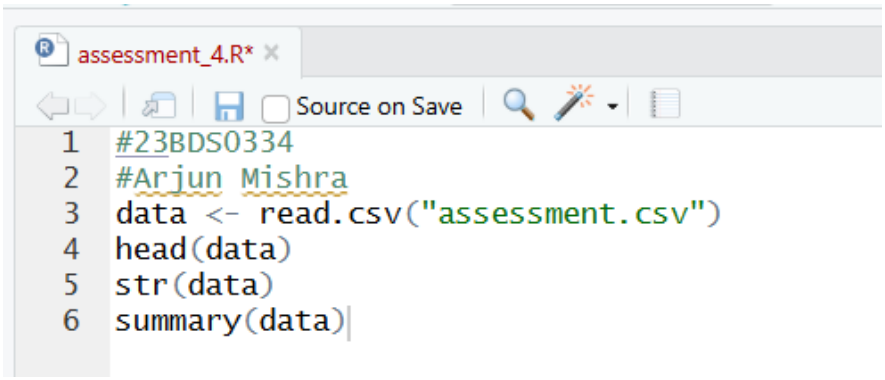
REG NO : 23BDS0334

1. **Load the dataset shared in your mail into the R programming environment and perform the following bivariate analysis and visualization tasks.**

AIM:

To load the dataset into the R programming environment and examine its structure.

CODE:



OUTPUT:

```
> #23BDS0334
> #Arjun Mishra
> data <- read.csv("assessment.csv")
> head(data)
  rownames Gunnel Time Fromlow Slope  Rw Amphiso Subst Pool Water Cobble
1        1      0  786     190    10 0.1       0     3    0     0      0
2        2      0  794     182    10 0.0       0     3    0     0      0
3        3      0  798     178    20 0.0       0     1    0     0      0
4        4      0  800     176    70 0.3       0     1    0     1      0
5        5      0  804     172    40 0.3       0     1    0     1      0
6        6      0  808     168    60 0.3       0     1    0     1      0
> str(data)
'data.frame':   1592 obs. of  11 variables:
 $ rownames: int  1 2 3 4 5 6 7 8 9 10 ...
 $ Gunnel  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Time    : int  786 794 798 800 804 808 814 817 821 826 ...
 $ Fromlow : int  190 182 178 176 172 168 162 159 155 150 ...
 $ Slope   : int  10 10 20 70 40 60 40 0 50 20 ...
 $ Rw      : num  0.1 0 0 0.3 0.3 0.3 0.3 0.9 0.5 0.1 ...
 $ Amphiso : int  0 0 0 0 0 0 0 1 0 0 ...
 $ Subst   : int  3 3 1 1 1 1 1 1 1 1 ...
 $ Pool    : int  0 0 0 0 0 0 0 1 0 0 ...
 $ Water   : int  0 0 0 1 1 1 1 1 1 0 ...
 $ Cobble  : int  0 0 0 0 0 0 0 0 0 0 ...
> summary(data)
    rownames         Gunnel             Time         Fromlow          Slope             Rw        
 Min.   :   1.0   Min.   :0.00000   Min.   :340.0   Min.   :  0.00   Min.   : 0.000   Min.   :0.0000  
 1st Qu.: 398.8   1st Qu.:0.00000   1st Qu.:623.0   1st Qu.: 71.75   1st Qu.: 0.000   1st Qu.:0.0000  
 Median : 796.5   Median :0.00000   Median :712.5   Median :158.00   Median : 0.000   Median :0.6000  
 Mean   : 796.5   Mean   :0.03015   Mean   :705.2   Mean   :149.41   Mean   : 9.504   Mean   :0.5271  
 3rd Qu.:1194.2   3rd Qu.:0.00000   3rd Qu.:810.0   3rd Qu.:224.25   3rd Qu.:10.000   3rd Qu.:1.0000  
 Max.   :1592.0   Max.   :1.00000   Max.   :983.0   Max.   :304.00   Max.   :90.000   Max.   :1.0000  
    Amphiso           Subst             Pool             Water            Cobble      
 Min.   :0.0000   Min.   : 1.000   Min.   :0.0000   Min.   :0.0000   Min.   :0.000  
 1st Qu.:0.0000   1st Qu.: 1.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.000  
 Median :0.0000   Median : 2.000   Median :0.0000   Median :1.0000   Median :0.000  
 Mean   :0.4642   Mean   : 3.106   Mean   :0.2023   Mean   :0.5088   Mean   :0.191  
 3rd Qu.:1.0000   3rd Qu.: 4.000   3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.:0.000  
 Max.   :4.0000   Max.   :13.000   Max.   :1.0000   Max.   :1.0000   Max.   :1.000  
> 
```

RESULT:

The assessment.csv dataset was successfully loaded into the R programming environment.

2. **Select two suitable categorical variables, create a contingency table, and display their relationship using:**

AIM:

To create a contingency table between two categorical variables and visualize their relationship using Base R and ggplot2 stacked bar plots.

CODE:

```
Source on Save
1  #23BDS0334
2  #Arjun Mishra
3  data$Amphiso <- factor(data$Amphiso)
4  data$Pool <- factor(data$Pool)
5  table(data$Amphiso, data$Pool)
```

```
R 4.5.2 · ~/
> #23BDS0334
> #Arjun Mishra
> data$Amphiso <- factor(data$Amphiso)
> data$Pool <- factor(data$Pool)
> table(data$Amphiso, data$Pool)

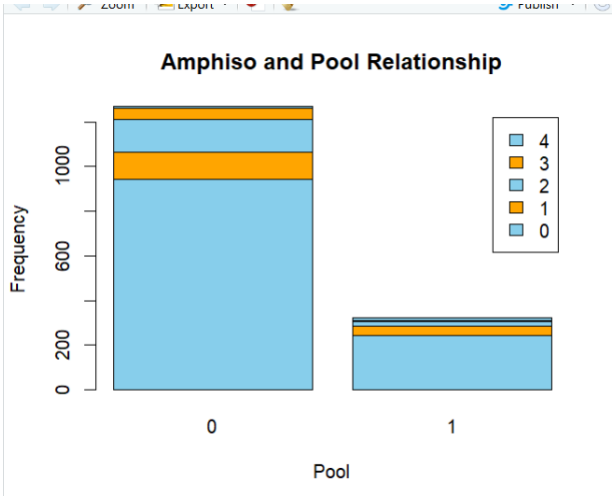
     0   1
  0 941 245
  1 125  40
  2 147  22
  3  49   3
  4   8  12
>
```

<u>Base R Stacked Bar Plot</u>

CODE:

```
assessment_4.R*
Source on Save
1  #23BDS0334
2  #Arjun Mishra
3  barplot(table(data$Amphiso, data$Pool),
4          beside=FALSE,
5          col=c("skyblue","orange"),
6          main="Amphiso and Pool Relationship",
7          xlab="Pool",
8          ylab="Frequency",
9          legend=TRUE)
```

OUTPUT:



ggplot2 Stacked Bar Plot

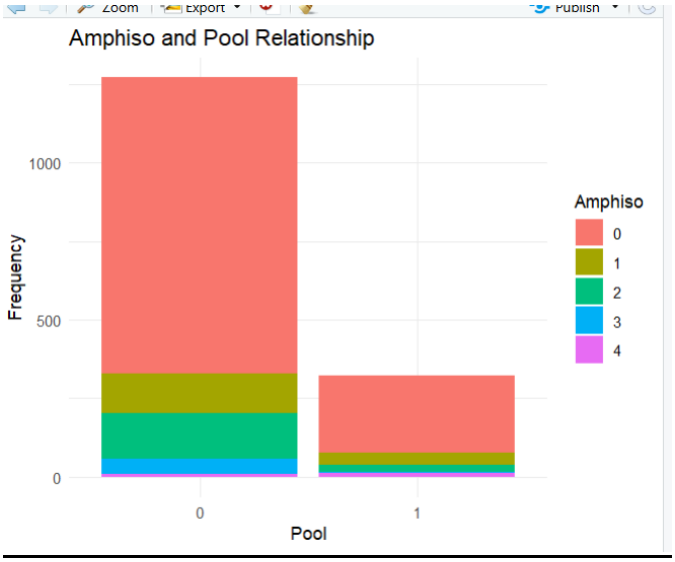
CODE:

assessment_4.R*

Source on Save

```
#23BDS0334
#Arjun Mishra
ggplot(data, aes(x=Pool, fill=Amphiso)) +
  geom_bar() +
  labs(title="Amphiso and Pool Relationship",
       x="Pool",
       y="Frequency",
       fill="Amphiso") +
  theme_minimal()
```

OUTPUT:



RESULT:

The relationship between the two categorical variables was successfully analyzed using a contingency table and stacked bar plots.

3. **Select two suitable categorical variables and create a grouped/side-by-side bar plot to compare their frequency distributions.**

AIM:

To compare the frequency distributions of two categorical variables using a grouped bar plot.

CODE:

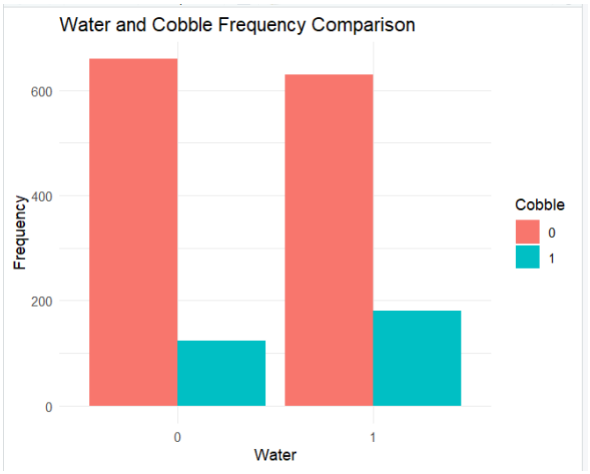
assessment_4.R*

Source on Save

```
#23BDS0334
#Arjun Mishra
data$Water <- factor(data$Water)
data$Cobble <- factor(data$Cobble)
ggplot(data, aes(x=Water, fill=Cobble)) +
  geom_bar(position="dodge") +
  labs(title="Water and Cobble Frequency Comparison",
       x="Water",
       y="Frequency",
       fill="Cobble") +
  theme_minimal()
```

OUTPUT:

```
> #23BDS0334
> #Arjun Mishra
> data$Water <- factor(data$Water)
> data$Cobble <- factor(data$Cobble)
> ggplot(data, aes(x=Water, fill=Cobble)) +
+   geom_bar(position="dodge") +
+   labs(title="Water and Cobble Frequency Comparison",
+        x="Water",
+        y="Frequency",
+        fill="Cobble") +
+   theme_minimal()
>
```



RESULT:

The frequency distributions of Water and Cobble were successfully compared using a grouped bar plot.

4. **Select two suitable numerical variables and generate a scatter plot to analyze the relationship between them.**

AIM:

To analyze the relationship between two numerical variables using a scatter plot.

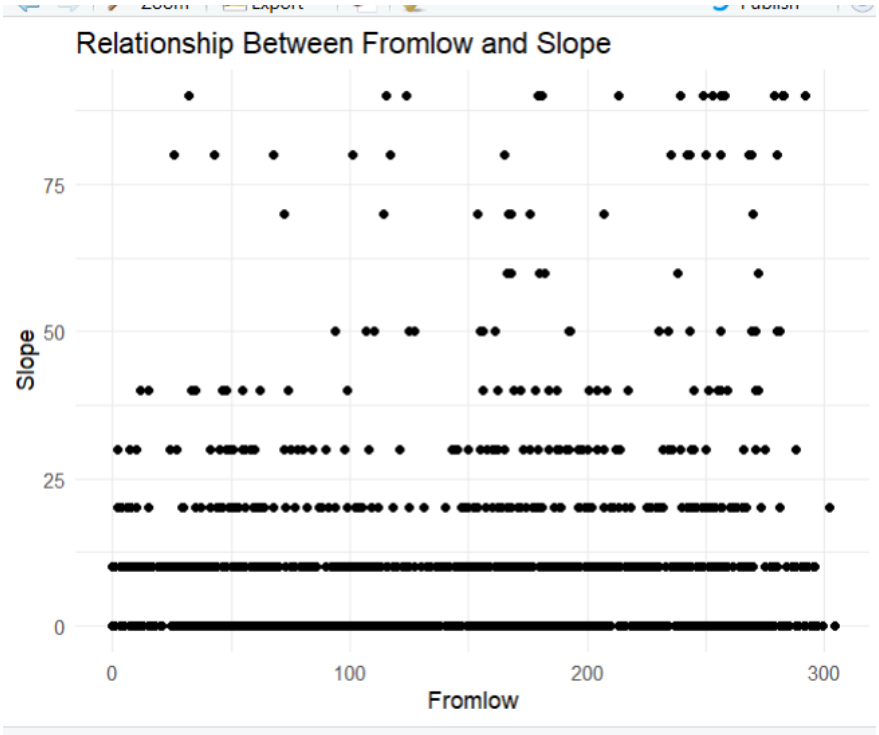
CODE:

assessment_4.R*

Source on Save

```
#23BDS0334
#Arjun Mishra
ggplot(data, aes(x=Fromlow, y=Slope)) +
  geom_point() +
  labs(title="Relationship Between Fromlow and Slope",
       x="Fromlow",
       y="Slope") +
  theme_minimal()
```

OUTPUT:



RESULT:

The relationship between the two numerical variables was successfully visualized using a scatter plot.

5. **Select two appropriate variables and construct a line plot to visualize their relationship or trend.**

AIM:

To visualize the trend between two numerical variables using a line plot.

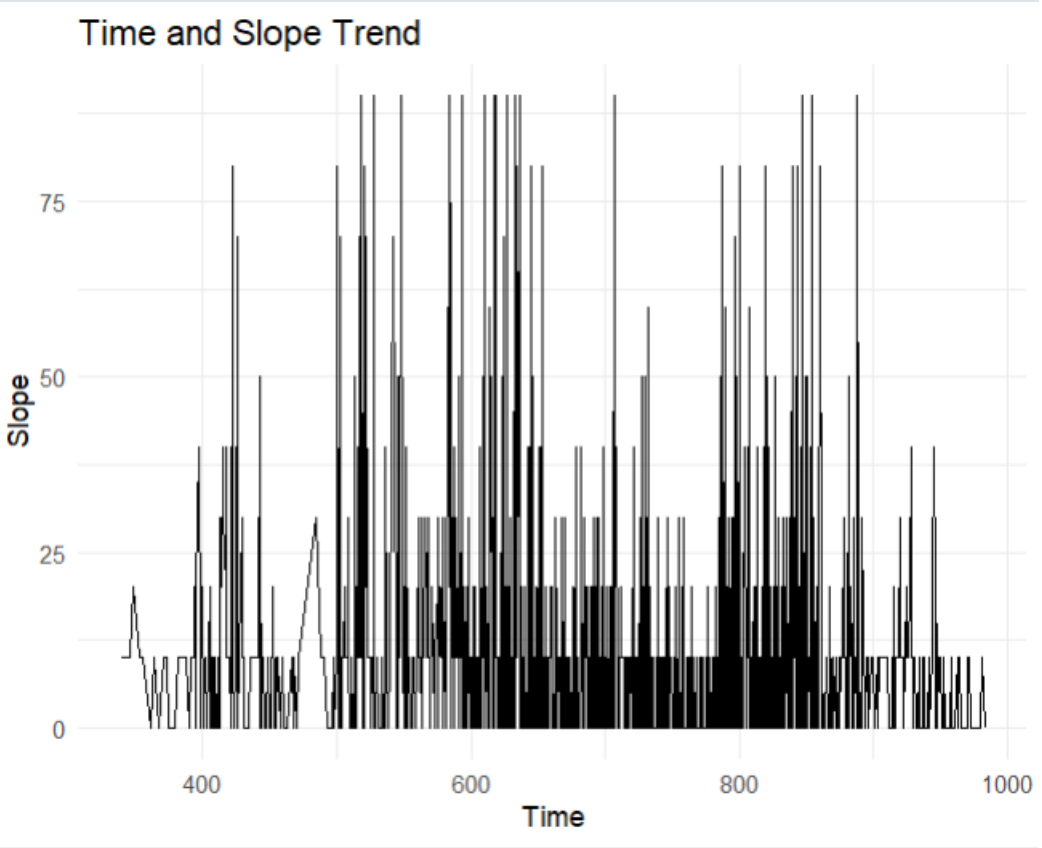
CODE:

assessment_4.R* ×

Source on Save

```
#23BDS0334
#Arjun Mishra
data <- data[order(data$Time),]
ggplot(data, aes(x=Time, y=Slope)) +
  geom_line() +
  labs(title="Time and Slope Trend",
       x="Time",
       y="Slope") +
  theme_minimal()
```

OUTPUT:



RESULT:

The trend between Time and Slope was successfully visualized using a line plot.

6. **Select suitable numerical variables and generate a heatmap to represent the relationship among the variables.**

AIM:

To visualize the correlation among selected numerical variables using a heatmap.

CODE:

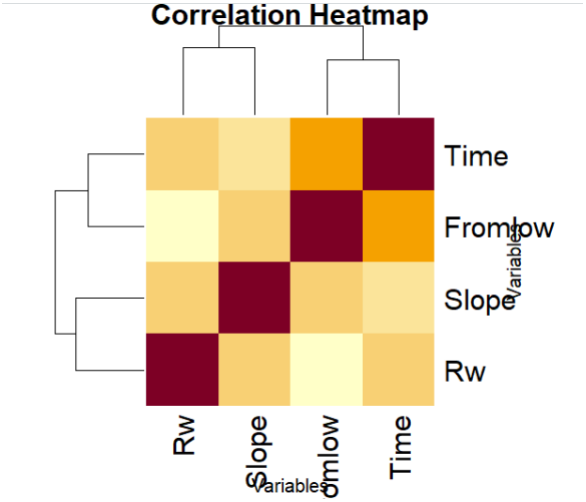
assessment_4.R*

Source on Save

```
#23BDS0334
#Arjun Mishra
x <- data[,c("Time","Fromlow","Slope","Rw")]
cor_matrix <- cor(x, use="complete.obs")
heatmap(cor_matrix,
        scale="none",
        main="Correlation Heatmap",
        xlab="Variables",
        ylab="Variables")
```

OUTPUT:

```
> #23BDS0334
> #Arjun Mishra
> x <- data[,c("Time","Fromlow","Slope","Rw")]
> cor_matrix <- cor(x, use="complete.obs")
> heatmap(cor_matrix,
+         scale="none",
+         main="Correlation Heatmap",
+         xlab="Variables",
+         ylab="Variables")
```



RESULT:

The relationships among the selected numerical variables were successfully represented using a heatmap.

7. **Select suitable variables and construct a bar chart to compare the values or frequencies between two variables.**

AIM:

To compare the average values of two numerical variables using a bar chart.

CODE:

assessment_4.R*

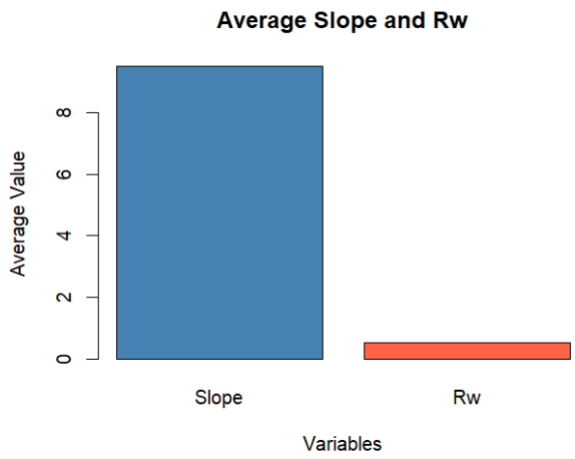
Source on Save

```
#23BDS0334
#Arjun Mishra
avg <- c(mean(data$Slope, na.rm=TRUE),
         mean(data$Rw, na.rm=TRUE))
barplot(avg,
        names.arg=c("Slope","Rw"),
        col=c("steelblue","tomato"),
        main="Average Slope and Rw",
        xlab="Variables",
        ylab="Average Value")
```

OUTPUT:

R 4.5.2 · ~/

```
> #23BDS0334
> #Arjun Mishra
> avg <- c(mean(data$Slope, na.rm=TRUE),
+          mean(data$Rw, na.rm=TRUE))
> barplot(avg,
+         names.arg=c("Slope","Rw"),
+         col=c("steelblue","tomato"),
+         main="Average Slope and Rw",
+         xlab="Variables",
+         ylab="Average Value")
>
```



RESULT:

The average values of the selected numerical variables were successfully compared using a bar chart.

8. **Select one numerical variable and one categorical variable and generate a density plot to compare the distribution of the numerical variable across different categories.**

AIM:

To compare the distribution of a numerical variable across different categories using a density plot.

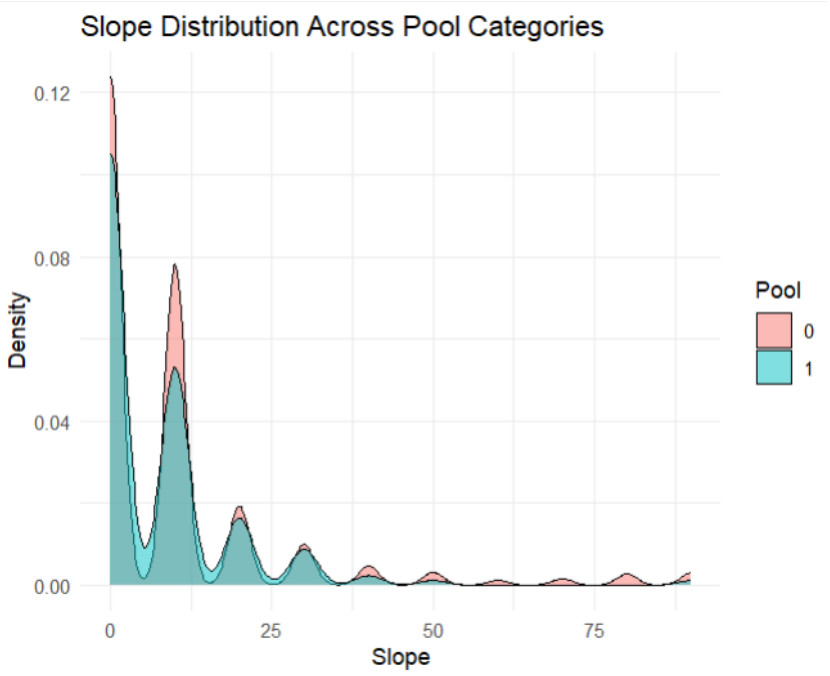
CODE:

assessment_4.R* ×

Source on Save

```r
#23BDS0334
#Arjun Mishra
data$Pool <- factor(data$Pool)
ggplot(data, aes(x=Slope, fill=Pool)) +
  geom_density(alpha=0.5) +
  labs(title="Slope Distribution Across Pool Categories",
       x="Slope",
       y="Density",
       fill="Pool") +
  theme_minimal()
```

OUTPUT:



RESULT:

The distribution of Slope across the Pool categories was successfully visualized using a density plot.

9. **Select one categorical variable and one numerical variable and create a box plot to compare the distribution of the numerical variable across different groups.**

AIM:

To compare the distribution of a numerical variable across different categorical groups using a box plot.

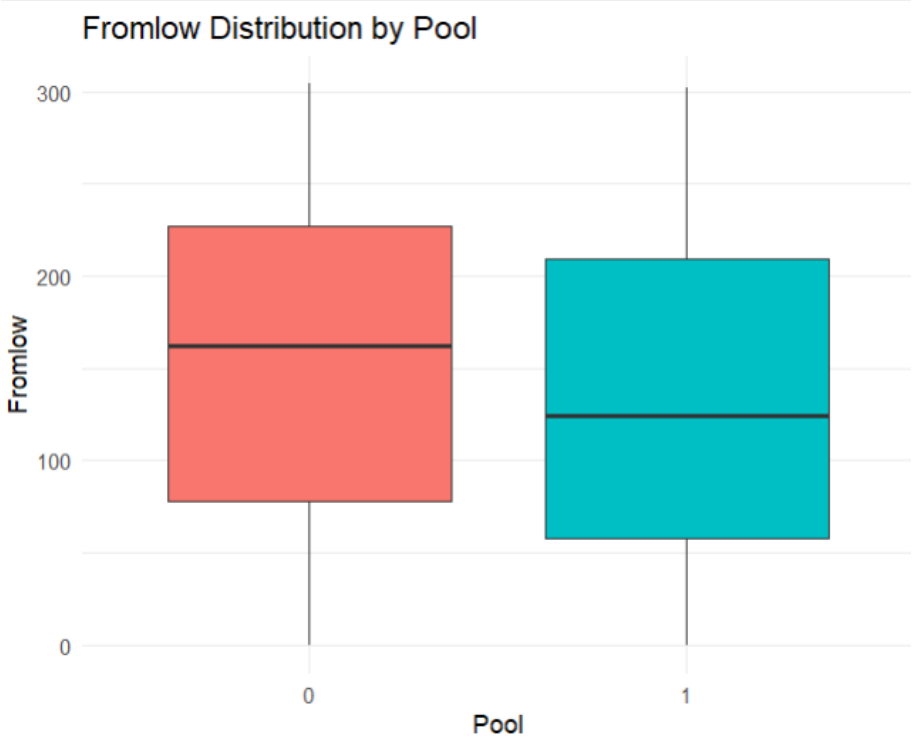
CODE:

assessment_4.R* ×

Source on Save

```
#23BDS0334
#Arjun Mishra
ggplot(data, aes(x=Pool, y=Fromlow, fill=Pool)) +
  geom_boxplot() +
  labs(title="Fromlow Distribution by Pool",
       x="Pool",
       y="Fromlow") +
  theme_minimal() +
  guides(fill="none")
```

OUTPUT:



RESULT:

The distribution of Fromlow across different Pool groups was successfully compared using a box plot.

10. Select one categorical variable and one numerical variable and construct a violin plot to visualize the distribution and density of the numerical variable across categories.

AIM:

To visualize the distribution and density of a numerical variable across different categories using a violin plot.

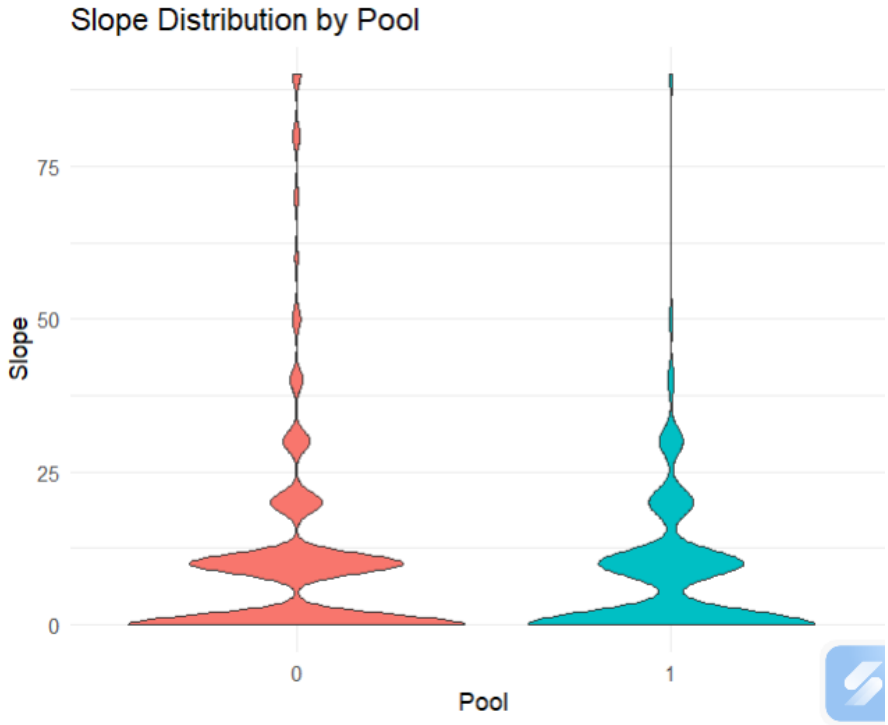
CODE:

assessment_4.R*

Source on Save

```
#23BDS0334
#Arjun Mishra
ggplot(data, aes(x=Pool, y=Slope, fill=Pool)) +
  geom_violin() +
  labs(title="Slope Distribution by Pool",
       x="Pool",
       y="Slope") +
  theme_minimal() +
  guides(fill="none")
```

OUTPUT:



RESULT:

The distribution and density of Slope across different Pool categories were successfully visualized using a violin plot.

11. Select suitable categorical and numerical variables and visualize their distribution by combining a violin plot and box plot in a single graphical representation.

AIM:

To visualize the distribution, density, median, quartiles, and outliers of a numerical variable across categories using a combined violin and box plot.

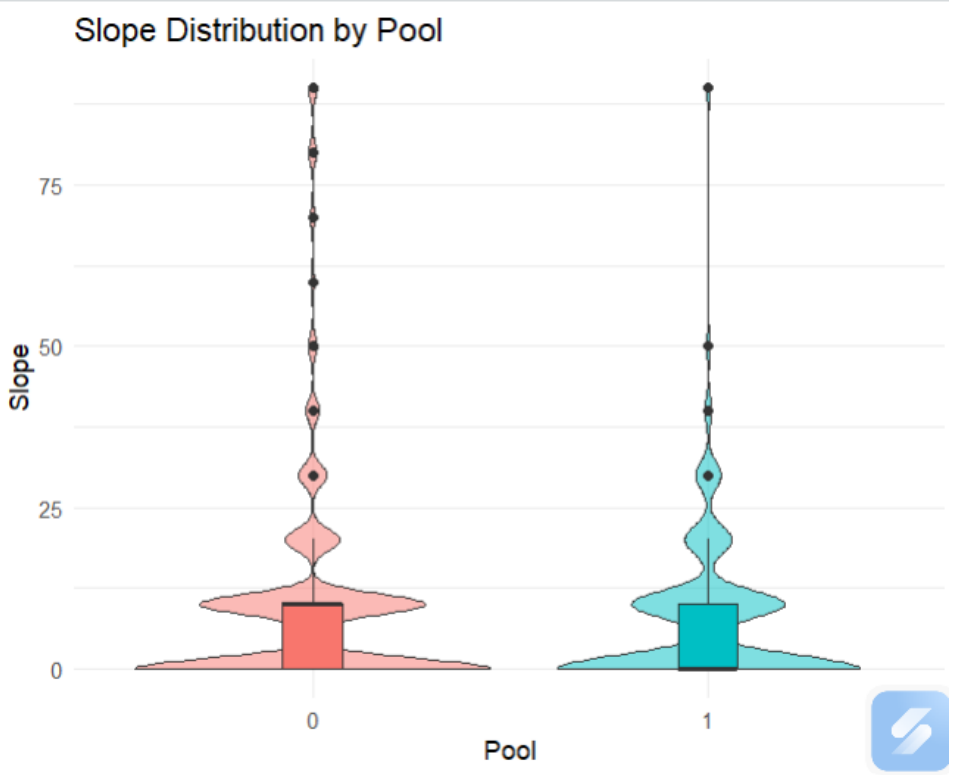
CODE:

assessment_4.R*

Source on Save

```
#23BDS0334
#Arjun Mishra
ggplot(data, aes(x=Pool, y=Slope, fill=Pool)) +
  geom_violin(alpha=0.5) +
  geom_boxplot(width=0.15) +
  labs(title="Slope Distribution by Pool",
       x="Pool",
       y="Slope") +
  theme_minimal() +
  guides(fill="none")
```

OUTPUT:



RESULT:

The distribution of Slope was successfully visualized using a combined violin and box plot.

12. Select suitable numerical variables, compute their correlation matrix, and generate a correlogram to study the strength and direction of relationships among the variables.

AIM:

To calculate the correlation matrix and generate a correlogram to study the strength and direction of relationships among numerical variables.

CODE:

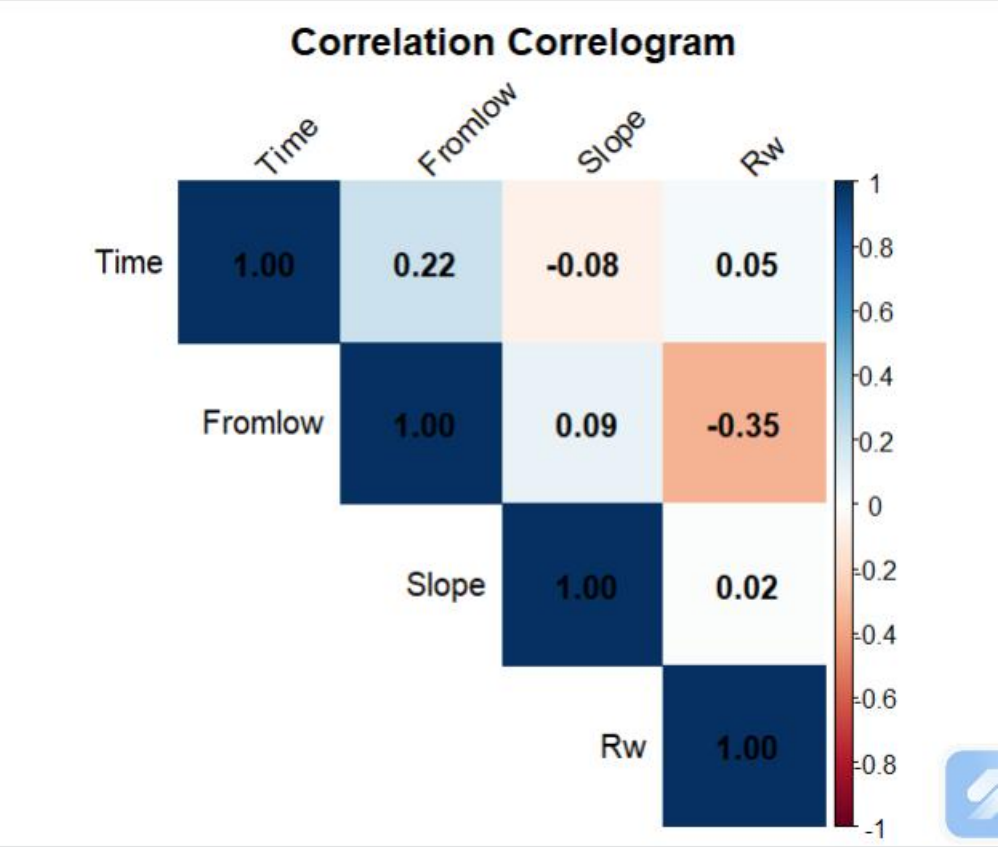
assessment_4.R* ×

Source on Save

```
#23BDS0334
#Arjun Mishra
x <- data[,c("Time","Fromlow","Slope","Rw")]
cor_matrix <- cor(x, use="complete.obs")
print(cor_matrix)
library(corrplot)
corrplot(cor_matrix,
         method="color",
         type="upper",
         addCoef.col="black",
         tl.col="black",
         tl.srt=45,
         title="Correlation Correlogram",
         mar=c(0,0,2,0))
```

OUTPUT:

```
> #23BDS0334
> #Arjun Mishra
> x <- data[,c("Time","Fromlow","Slope","Rw")]
> cor_matrix <- cor(x, use="complete.obs")
> print(cor_matrix)
                Time    Fromlow       Slope          Rw
Time      1.00000000  0.2155932 -0.07692201  0.04781209
Fromlow   0.21559316  1.0000000  0.09058570 -0.34861156
Slope    -0.07692201  0.0905857  1.00000000  0.01956647
Rw        0.04781209 -0.3486116  0.01956647  1.00000000
> library(corrplot)
corrplot 0.95 loaded
Warning message:
package 'corrplot' was built under R version 4.5.3
> corrplot(cor_matrix,
+          method="color",
+          type="upper",
+          addCoef.col="black",
+          tl.col="black",
+          tl.srt=45,
+          title="Correlation Correlogram",
+          mar=c(0,0,2,0))
>
```



RESULT:

The strength and direction of relationships among the numerical variables were successfully analyzed using a correlation matrix and correlogram.

13. Provide suitable titles, axis labels, legends, and graphical parameters for all the plots generated.

AIM:

To apply suitable titles, axis labels, legends, and graphical parameters to the generated plots.

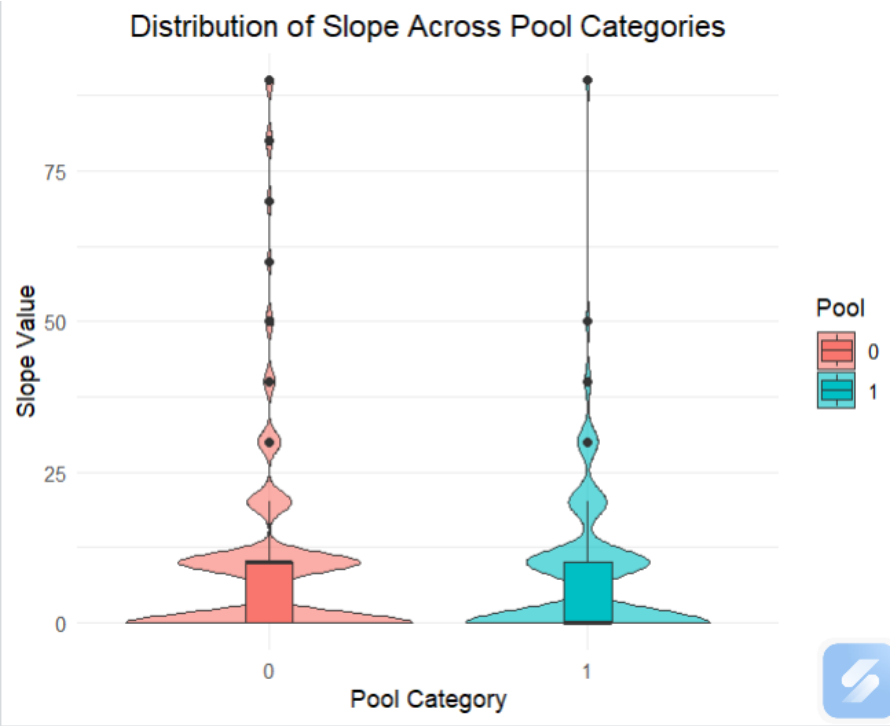
CODE:

assessment_4.R*

Source on Save

```
#23BDS0334
#Arjun Mishra
ggplot(data, aes(x=Pool, y=Slope, fill=Pool)) +
  geom_violin(alpha=0.6) +
  geom_boxplot(width=0.15) +
  labs(title="Distribution of Slope Across Pool Categories",
       x="Pool Category",
       y="Slope Value",
       fill="Pool") +
  theme_minimal() +
  theme(plot.title=element_text(hjust=0.5),
        axis.title=element_text(size=11),
        legend.position="right")
```

OUTPUT:



RESULT:

Suitable titles, axis labels, legends, and graphical parameters were successfully applied to the visualization.